Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two cars are moving in the same direction with the same speed $30\ \mathrm{km/hr}$. They are separated by a distance of $5\ \mathrm{km}$. The speed of a car moving in the opposite direction if it meets these two cars at an interval of 4 minutes, will be

(A) $40\ \mathrm{km/hr}$

(B) $45\ \mathrm{km/hr}$

(C) $30\ \mathrm{km/hr}$

(D) $15\ \mathrm{km/hr}$

Q.2

A thief is running away on a straight road on a jeep moving with a speed of $9\ \mathrm{m/s}$. A police man chases him on a motor cycle moving at a speed of $10\ \mathrm{m/s}$. If the instantaneous separation of jeep from the motor cycle is $100\ \mathrm{m}$, how long time (in sec) will it take for the police man to catch the thief?

(A) 110

(B) 120

(C) 130

(D) 100

Q.3

A particle is moving eastwards with a velocity of $5\ \mathrm{ms}^{-1}$. In 10 seconds the velocity changes to 5 ms^{-1} northwards. The average acceleration in this time is

(A) $\frac{1}{2}\ \mathrm{ms}^{-2}$ towards north

(B) $\frac{1}{\sqrt{2}}\ \mathrm{ms}^{-2}$ towards north - east

(C) $\frac{1}{\sqrt{2}}\ \mathrm{ms}^{-2}$ towards north - west

(D) zero

Q.4

Two trains, each $40\ \mathrm{m}$ long are travelling in opposite direction with equal velocity $20\ \mathrm{m/s}$. The time of crossing is

(A) $1\ \mathrm{s}$

(B) $2\ \mathrm{s}$

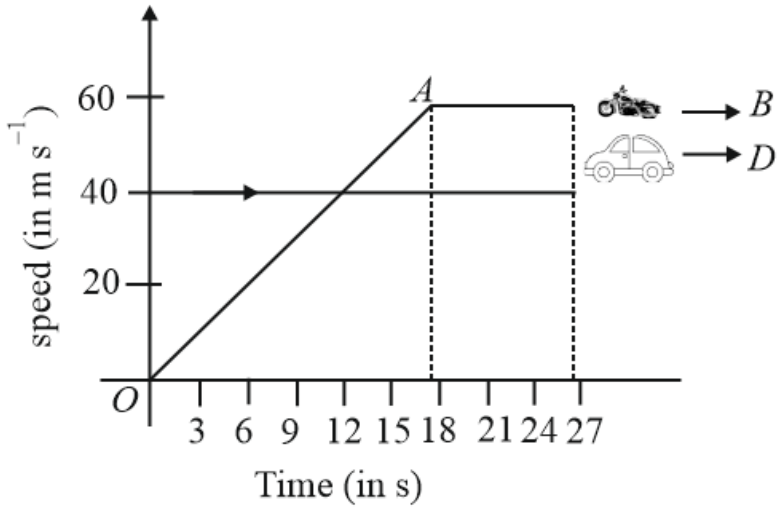
(C) $3\ \mathrm{s}$

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(D) Zero

Q.5

At the instant a motor bike starts from rest in a given direction, a car overtakes the motor bike, both moving in the same direction. The speed-time graphs for motor bike and car are represented by OAB and CD respectively. Then



(A) At $t = 18$ s the motor bike and car are 280m apart

(B) At $t = 18$ s the motor bike and car are 720m apart

(C) The relative distance between motor bike and car reduces to zero at $t = 27$ s and both are 1080m far from origin

(D) The relative distance between motor bike and car always remains same

Q.6

A train of $150\ \text{m}$ length is going towards north direction at a speed of $10\ \text{ms}^{-1}$. A parrot flies at a speed of $5\ \text{ms}^{-1}$ towards south direction parallel to the railway track. The time taken by the parrot to cross the train is equal to

(A) $12\ \text{s}$

(B) $8\ \text{s}$

(C) $15\ \text{s}$

(D) $10\ \text{s}$

Q.7

A boat which has a speed of $5\ \text{km/hr}$ in still water crosses a river of width $1\ \text{km}$ along the shortest possible path in 15 minutes. Then what is the velocity of the river water in km/hr ?

(A) 1

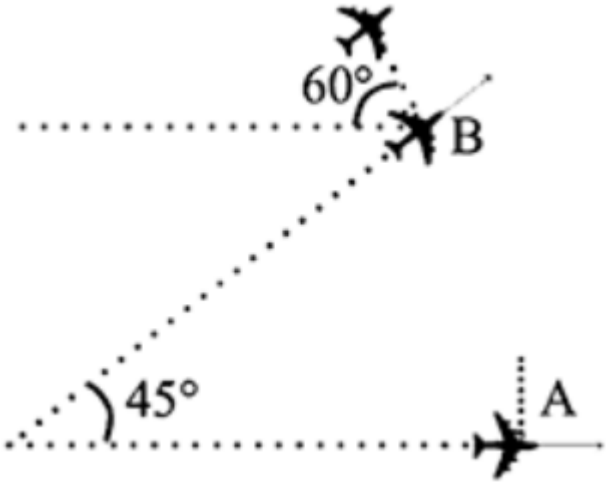
(B) 2

(C) 4

(D) 3

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Q.8



Passengers in the jet transport A flying east at a speed of 800 kmh^{-1} observe a second jet plane B that passes under the transport in horizontal flight. Although the nose of B is pointed in the 45° north east direction, plane B appears to the passengers in A to be moving away from the transport at the 60° angle as shown. Then what is the true velocity (in $\mathrm{km/h}$) of B ?

(A) 727

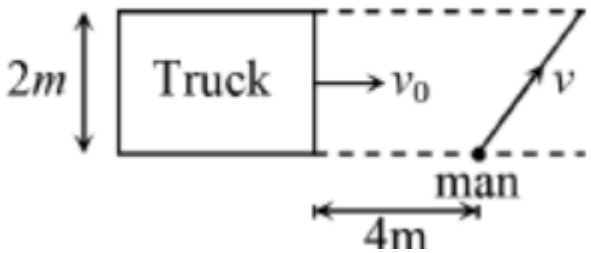
(B) 737

(C) 707

(D) 717

Q.9

A $2\ \mathrm{m}$ wide truck is moving with a uniform speed $v_0 = 8\ \mathrm{m/s}$ along a straight horizontal road. A pedestrian starts to cross the road with a uniform speed v when the truck is $4\ \mathrm{m}$ away from him. The minimum value of v so that he can cross the road safely is



(A) $2.62\ \mathrm{m/s}$

(B) $4.6\ \mathrm{m/s}$

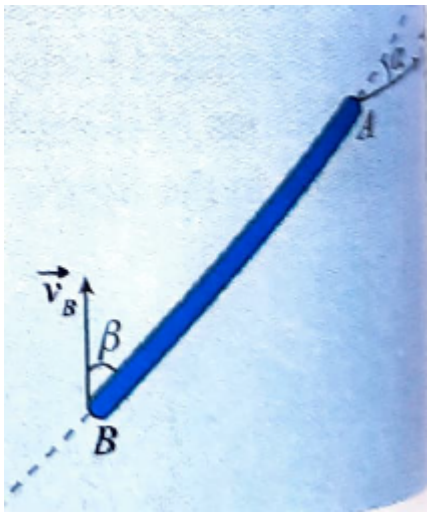
(C) $3.57\ \mathrm{m/s}$

(D) $1.414\ \mathrm{m/s}$

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Q.10

The ends A and B of a rod of length l have velocities of magnitudes $|\vec{v}_A| = v$ and $|\vec{v}_B| = 2v$, respectively. If the inclination of $\vec{v}_A$ relative to the rod is α, find the:
angular velocity of the rod.



(A) $\frac{\left(\sin\alpha+\sqrt{3+\sin^2\alpha}\right)v}{l}$

(B) $\frac{\left(\tan\alpha+\sqrt{3}+\sin^2\alpha\right)v}{l}$

(C) $\frac{\left(\sin\alpha+\sqrt{4}+\sin^2\alpha\right)v}{l}$

(D) $\frac{\left(\sin\alpha+\sqrt{5}+\sin^2\alpha\right)v}{l}$

Q.11

A rod of length l resting on a wall and the floor. Its lower end A is pulled towards right with a constant velocity v. Find the velocity of the other end B downward when the rod makes an angle θ with the vertical. Also, find the angular velocity of the rod.

(A) $\frac{v}{l\cos\theta}$

(B) $l\cos\theta$

(C) $\frac{v}{l\sin\theta}$

(D) $\frac{v}{l^2\cos\theta}$

Q.12

The driver of a train moving at a speed v_1 sights another train at a distance d ahead of him, moving in the same direction with a slower speed v_2. He immediately applies brakes to achieve a constant retardation a. There will be no collision if d is greater than

(A) $\frac{(v_1-v_2)^2}{a}$

(B) $\frac{\left(v_1^2-v_2^2\right)}{a}$

(C) $\frac{(v_1-v_2)^2}{2a}$

(D) $\frac{\left(v_1^2-v_2^2\right)}{2a}$

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Q.13

The driver of train A moving at a speed of $30\ \text{ms}^{-1}$ sights another train B moving on the same track at a speed of $10\ \text{ms}^{-1}$ in the same direction. He immediately applies brakes and achieves a uniform retardation of $2\ \text{ms}^{-2}$. To avoid collision, what must be the minimum distance between trains A and B when the driver of A sights B ?

(A) 120 m

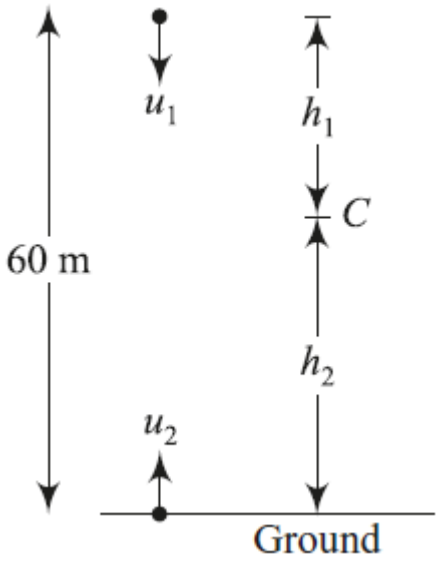
(B) 100 m

(C) 180 m

(D) 130 m

Q.14

From the top of a tower $60\ \text{m}$ tall, a body is thrown vertically down with a velocity of $10\ \text{ms}^{-1}$. At the same time, another body is thrown vertically upward from the ground with a velocity of $20\ \text{ms}^{-1}$. (a) After how long will the two bodies meet? (b) At what height above the ground do they meet? Take $g = 10\ \text{ms}^{-2}$.



(A) 2 s, 20 m

(B) 4 s, 30 m

(C) 5 s, 20 m

(D) 7 s, 30 m

Q.15

A police van moving on a highway with a speed of $10\ \text{ms}^{-1}$ fires a bullet at a thief's car speeding away in the same direction with a speed of 30 ms^{-1}. If the muzzle speed of the bullet is $140\ \text{ms}^{-1}$, with what speed will the bullet hit the thief's car?

(A) 160 ms^{-1}

(B) 170 ms^{-1}

(C) 120 ms^{-1}

(D) 150 ms^{-1}

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Q.1 Answer:

$45\ \mathrm{km/hr}$

Solution:

The two car (say A and B) are moving with same velocity, the relative velocity of one(say B) with respect to the other A, $\overrightarrow{\mathrm{v}}_{\mathrm{BA}} = \overrightarrow{\mathrm{v}}_{\mathrm{B}} - \overrightarrow{\mathrm{v}}_{\mathrm{A}} = \mathrm{v} - \mathrm{v} = 0$

So the relative separation between them $(= 5\ \mathrm{km})$ **always remains the same. Now if the velocity of car (say C) moving in opposite**

direction to A **and** B**, is** $\overrightarrow{\mathrm{v}}_{\mathrm{C}}$ **relative to ground then the velocity of car** C **relative to** A **and** B **will be** $\vec{v}_{\text{rel.}} = \vec{v}_c - \vec{v}$

But as $\overrightarrow{\mathrm{v}}$ **is opposite to** $\mathrm{v_c}$ **So, the time taken by it to cross the cars** A **and** B

$$\mathrm{t} = \frac{\mathrm{d}}{\mathrm{v_{rel.}}} \Rightarrow \frac{4}{60} = \frac{5}{\mathrm{v_C} + 30} \Rightarrow \mathrm{v_c} = 45\ \mathrm{km/hr}$$

Q.2 Answer:

100

Solution:

Relative speed of police with respect to thief $= 10 - 9 = 1\ \mathrm{m/s}$

Instantaneous separation $= 100\ \mathrm{m}$

Time $= \frac{\text{Distance}}{\text{Velocity}} = \frac{100}{1} = 100\mathrm{sec}$

Q.3 Answer:

$\frac{1}{\sqrt{2}}\ \mathrm{ms}^{-2}$ towards north - west

Solution:

$$\text{Average acceleration } = \frac{\text{change in velocity}}{\text{time interval}}$$

$= \frac{\Delta \overrightarrow{\mathrm{v}}}{\mathrm{t}}$

$\overrightarrow{\mathrm{v_1}} = 5\hat{\mathrm{i}}, \overrightarrow{\mathrm{v_2}} = 5\hat{\mathrm{j}}$

$\Delta \overrightarrow{\mathrm{v}} = \left(\overrightarrow{\mathrm{v}}_2 - \overrightarrow{\mathrm{v}}_1\right)$

$= \sqrt{\mathrm{v_1^2} + \mathrm{v_2^2} + 2\mathrm{v_1 v_2} \cos 90}$

$= \sqrt{5^2 + 5^2 + 0}$

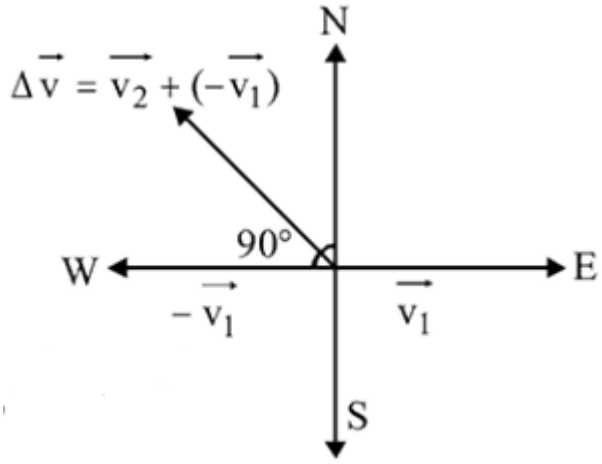
$[\mathrm{As}\ |\mathrm{v_1}| = |\mathrm{v_2}| = 5\ \mathrm{m/s}] = 5\sqrt{2}\ \mathrm{m/s}$

Avg. acc. $= \frac{\Delta \overrightarrow{\mathrm{v}}}{\mathrm{t}} = \frac{5\sqrt{2}}{10} = \frac{1}{\sqrt{2}}\ \mathrm{m/s^2}$

$\Rightarrow \tan\theta = \frac{5}{-5} = -1$

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which means θ is in the second quadrant. (towards north-west)



Q.4 Answer:

2 s

Solution:

Time $= \frac{\text{total length}}{\text{relative velocity}} = \frac{X+X}{20+20} = \frac{2X}{40}$ s

$= 2$ s

Q.5 Answer:

The relative distance between motor bike and car reduces to zero at $t = 27$ s and both are 1080m far from origin

Solution:

Distance travelled by motor bike at $t = 18$**s**

$s_{bike} = s_1 = ut + \frac{1}{2}at^2u = 0 \Rightarrow \quad s_1 = \frac{1}{2}at = \frac{1}{2} \times 18 \times 60 = 540m$

Distance travelled by car moving with uniform velocity at $t = 18$**s**

$s_{car} = s_2 = ut = 18 \times 60 = 720m \quad (a = 0)$

Therefore, separation between them at $t = 18$**s is 180m.**

Let, separation between them decreases to zero at time t **beyond 18s.**

Hence, Displacement of bike in 27 sec is $s_{bike} = 540 + 60t = 540 + 60 \times 9 = 1080m$

and displacement of car in 27 sec is $s_{car} = 40t = 40 \times 27 = 1080m$

$s_{car} = s_{bike}$

Q.6 Answer:

10 s

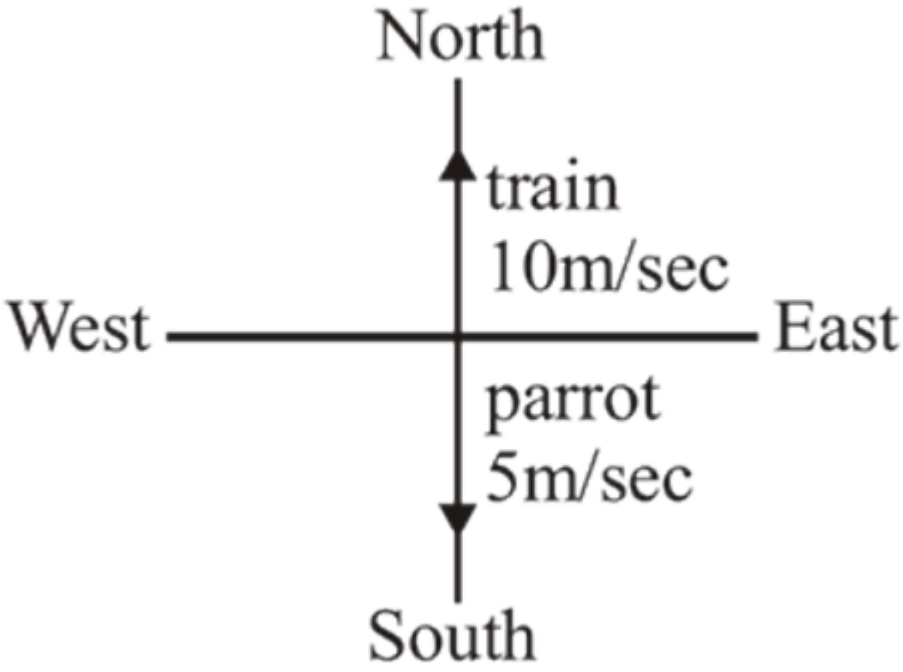
Solution:

So by figure the velocity of parrot w.r. t. train is $= 5 - (-10) = 15$ m/sec

so time taken to cross the train is

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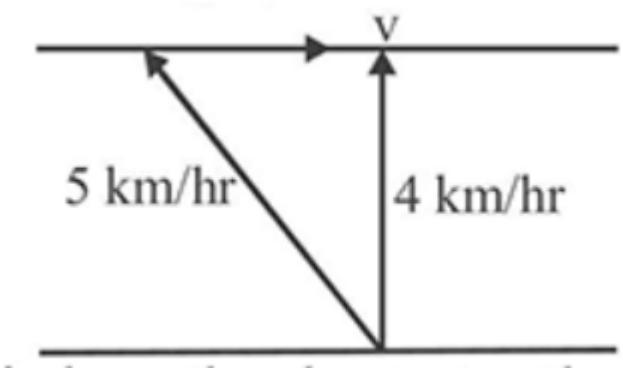
$= \frac{\text{length of train}}{\text{relative velocity}} = \frac{150}{15} = 10\text{sec}$



Q.7 Answer:

3

Solution:



Speed along the shortest path $= \frac{1}{15/60} = 4\ \text{km/hr}$

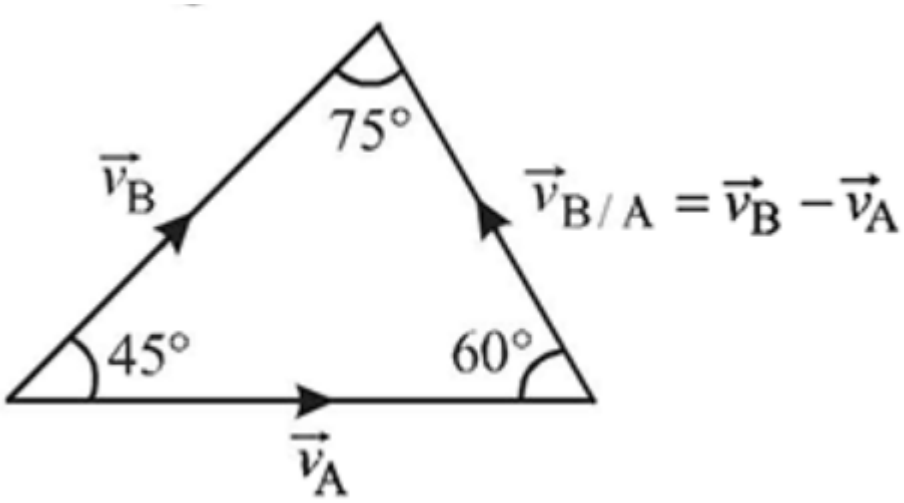
Speed of water $v = \sqrt{5^2 - 4^2} = 3\ \text{km/hr}$

Q.8 Answer:

717

Solution:

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According to law of sines or Lami's Theorem

$\Rightarrow \frac{v_A}{\sin 75^\circ} = \frac{v_B}{\sin 60^\circ} = \frac{v_{B/A}}{\sin 45^\circ}$

$\Rightarrow v_B = 717 \text{kmh}^{-1}$

Q.9 Answer:

3.57 m/s

Solution:

For safe crossing, the condition is that the man must cross the road by the time the truck covers the distance $4 + AC$ **or** $4 + 2\cot\theta$

$\therefore$

$\frac{4+2\cot\theta}{8} = \frac{2/\sin\theta}{v}$

or $v = \frac{8}{2\sin\theta + \cos\theta}$

For minimum v, $\frac{dv}{d\theta} = 0 \quad \Rightarrow \quad \tan\theta = 2$

From equation (i), $v_{\min} = \frac{8}{\sqrt{5}} = 3.57 \text{ m/s}$

$\therefore$

Q.10 Answer:

$\frac{\left(\sin\alpha + \sqrt{3+\sin^2\alpha}\right)v}{l}$

Solution:

(a) We know in a rigid body the velocities of two points along the line joining two should remain same.

i.e., $(v_A)_{||} = (v_B)_{||}$

Hence $v\cos\alpha = 2v\cos\beta$

$\cos\beta = \frac{1}{2}\cos\alpha$ **or** $\beta = \cos^{-1}\left(\frac{1}{2}\cos\alpha\right)$

(b) Angular velocity of the rod, $\omega = \frac{|(\vec{v}_{AB})_\perp|}{l} = \frac{|(\vec{v}_A - \vec{v}_B)_\perp|}{l}$

$|\vec{v}_{AB}| = |\vec{v}_A - \vec{v}_B| = v\sin\alpha - (-2v\sin\beta)$

$= v(\sin\alpha + 2\sin\beta)$

$\Rightarrow \omega = \frac{(\sin\alpha + 2\sin\beta)v}{l}$ **(in clockwise direction)**

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We have calculated $\cos\beta = \frac{\cos\alpha}{2}$. Thus,

$\sin\beta = \sqrt{1-\cos^2\beta} \Rightarrow \sin\beta = \frac{\sqrt{3+\sin^2\alpha}}{2}$

Hence $\omega = \frac{\left(\sin\alpha+\sqrt{3+\sin^2\alpha}\right)v}{l}$ (in clockwise direction)

Q.11 Answer:

$\frac{v}{l\cos\theta}$

Solution:

As rod is a rigid body, the distance between point A and point B should remain unchanged. Hence the relative velocity of ends of rod along its length should be zero.

It means the velocities of ends of rod along its length must be equal.

$$v_A \sin\theta = v_B \cos\theta \Rightarrow v_B = v_A \frac{\sin\theta}{\cos\theta} = v\tan\theta$$

or $\quad v_B = v\tan\theta \quad ...(\text{i})$

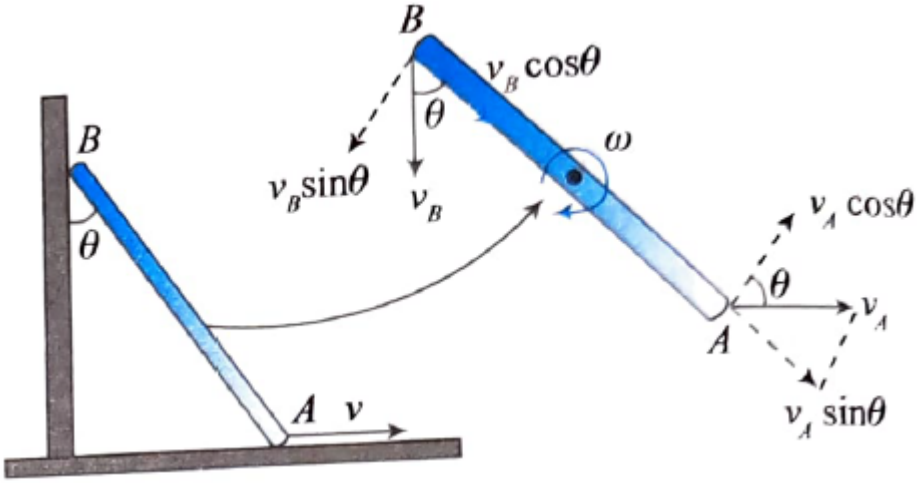
Then, $\omega_{BA} = \frac{v_{BA_\perp}}{l} = \frac{|v_A\cos\theta-(-v_B\sin\theta)|}{l}$

$\Rightarrow \quad \omega_{BA} = \frac{v_A\cos\theta+v_B\sin\theta}{l}$ **...(ii)**

Using Eqs. (i) and (ii),

$\omega_{BA} = \frac{v}{l\cos\theta}$

This is also angular velocity of rigid body about its centre of mass.



Q.12 Answer:

$\frac{(v_1-v_2)^2}{2a}$

Solution:

The two trains will not collide if the initial relative velocity $u = (v_1 - v_2)$ reduces to zero relative velocity $(v = 0)$ in a minimum distance $S = d_{\min}$ under a retardation $(-a)$. Using $v^2 - u^2 = 2aS$, we have

$$0 - (v_1 - v_2)^2 = 2(-a)d_{\min}$$

$$\Rightarrow \quad d_{\min} = \frac{(v_1 - v_2)^2}{2a}$$

So the correct choice is (c).

Q.13 Answer:

100 m

Solution:

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Relative initial velocity A w.r.t. B is,

$u_{AB} = u_A - u_B = 30 - 10 = 20 \text{ ms}^{-1}$

Relative retardation of A w.r.t B is

$a_{AB} = a_A - a_B = -2 - 0 = -2 \text{ ms}^{-2}$

To avoid collision, the relative final velocity of A w.r.t. $B = v_{AB}$ must be zero. Minimum relative displacement of A

w.r.t. B is s_{AB} which is found by using the relation $v^2 - u^2 = 2$ as which gives

$0 - (20)^2 = 2 \times (-2) s_{AB} \Rightarrow s_{AB} = 100 \text{ m}$

Q.14 Answer:

2 s, 20 m

Solution:

(a) Suppose the bodies meet at C and let t be the time at which they meet.

For body 1: $s = -h_1, u_1 = -10 \text{ ms}^{-1}, a = -10 \text{ ms}^{-2}$

$\therefore \quad -h_1 = -10t + \frac{1}{2} \times (-10) t^2$

which gives $h_1 = 5t(t+2)$ **(1)**

For body 2: $s = +h_2, u_2 = +20 \text{ ms}^{-1},$

$a = -10 \text{ ms}^{-2}$

$\therefore \quad h_2 = 20t - 5t^2 = 5t(-t+4) \quad (2)$

Adding (1) and (2), $h_1 + h_2 = 30t$ **or** $60 = 30t \Rightarrow \quad t = 2 \text{ s}.$

(b) Using $t = 2 \text{ s}$ **in Eq. (2),** $h_2 = 5 \times 2(4-2) = 20 \text{ m}$

Q.15 Answer:

120 ms^{-1}

Solution:

Speed of police van is $v_V = 10 \text{ ms}^{-1}$, speed of thief's car is $v_C = 30 \text{ ms}^{-1}$. Relative velocity of bullet with respect to van is $v_{BV} = 140 \text{ ms}^{-1}$. Let v_B be the velocity of the bullet. $v_{BV} = v_B - v_V$. Hence $v_B = v_{BV} + v_V = 140 + 10 = 150 \text{ ms}^{-1}$. The bullet will hit the car with a speed

$v_{BC} = v_B - v_C = 150 - 30 = 120 \text{ ms}^{-1}$

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